

ALPHABETS, STRINGS AND LANGUAGES -BASIC MATHEMATICAL NOTATIONS

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LEARNING OBJECTIVE

- To construct finite automata for any given pattern and find its equivalent regular expressions
 - Before learning FA learn the Basic Mathematical Notation and Techniques

ALPHABET

- An alphabet is a finite, non-empty set of symbols. It is denoted by Σ .

$$^{26} \Sigma = \{a, b, c \dots z\}$$

Examples

- $\Sigma = \{a, b\}$ $\rightarrow$ alphabet of 2 symbols a and b
- $\Sigma = \{0, 1, 2\}$ $\rightarrow$ an alphabet of 3 symbols 0, 1 and 2

STRING

- A string (or) word is a finite sequence of symbols chosen from some alphabet Σ .

Examples

- $\Sigma = \{a, b\}$
 - Strings $\rightarrow$ abab, aabba, aaabba ... a, b, ab bb ...
- $\Sigma = \{a\}$
 - Strings $\rightarrow$ a, aa, aaa ...
- Notations
 - **a, b, c** $\rightarrow$ elements of Σ
 - **u, v, w** $\rightarrow$ string names.

STRING

- $\Sigma = \{0,1\}$
- Strings??

0, 1, 00, 11, 010, 101, ...

00000

11111

OPERATIONS ON STRINGS

1. LENGTH OF A STRING

- The **length** of a string **x** is the number of symbols contained in the string **x**, denoted by **|x|**.
- Example

$$| \text{string} | = 6$$

$$| \text{UCS1503} | = 7$$

$$| 101001 | = 6$$

$$| \epsilon | = 0$$

2. EMPTY (OR) NULL STRING

- The empty string is the string with zero occurrences of symbols or the length of a string is zero.
- It is denoted by ε or λ .
- $|\varepsilon| = 0 = |\lambda|$

$$\Sigma = \{a, b\}$$

$a, b, aa, bb, \dots$

ε
null string

(ε)

3. CONCATENATION OF STRING

- Let x and y be strings. Then xy denotes the concatenation of x and y , that is, the string formed by making a copy of x and following it by a copy of y .

$x = a_1 a_2 a_3 \dots a_m$

$y = b_1 b_2 b_3 \dots b_n$

then $xy = a_1 a_2 a_3 \dots a_m \underline{b_1} \underline{b_2} b_3 \dots b_n$

- The length of the string is $m+n$

Examples

$x = \underline{010} \ y = 1$

$xy = \underline{0101} \ yx = \underline{1010}$

$x = \text{CS} \ y = 6503$

$xy = \text{CS6503}$

$x = 010$
 $y = \epsilon$

$xy = 010$

$yx = 010$

$w\epsilon = \epsilon w = w$

Empty string is the identity element for concatenation operator ie. $w\epsilon$ = ϵw = w

3. CONCATENATION OF STRING

- $x = \text{apple}$

- $y = \text{an}$

- $xy = ?$

applean

- $yx = ?$

anapple

- $yy = ?$

anan

- $xyx = ?$ *appleanapple*

- $x = \varepsilon$ *null string*

- $y = \{a, b\}$

- $z = \phi$ *→ no string*

- $xy = ?$

$\varepsilon \{a, b\} = \{a, b\}$

- $yz = ?$

$\{a, b\} \cdot \phi = \phi$

4. REVERSE OF A STRING

- The reverse of a string is obtained by writing the symbols in reverse order.

Let w be a string. Then its reverse is w^R

ie. $w = a_1 a_2 a_3 \dots a_m$

$w^R = a_m \dots a_2 a_1$

Example

Let $u = 0101011$

$u^R =$ 1101010

4. REVERSE OF A STRING

- $x = \text{apple}$

- $y = \text{an}$

- $x^R = ?$

elppa

- $y^R = ?$

na

- $yy^R = ?$

anna

- $xy^Rx^R = ?$ *apple na elppa*

5. POWERS OF AN ALPHABET

- Let Σ be an alphabet.
- Σ^* denotes the set of all strings over the alphabet Σ .
- Σ^m denotes the set of all strings over the alphabet Σ of length m .

Example

If $\Sigma = \{0, 1\}$ then

- $\Sigma^0 = \{\epsilon\}$ empty string
- $\Sigma^1 = \{0, 1\}$ set of all strings of length one over $\Sigma = \{0, 1\}$
- $\Sigma^2 = \{00, 01, 10, 11\}$ set of all strings of length two over $\Sigma = \{0, 1\}$

$$\Sigma \cdot \Sigma = \{0, 1\} \cdot \{0, 1\} = \{00, 01, 10, 11\}$$

5. POWERS OF AN ALPHABET

- $\Sigma = \{a, b\}$
- $\Sigma^0 = \{\epsilon\}$
- $\Sigma^1 = \{a, b\}$
- $\underline{\Sigma^2} = \Sigma^1 \cdot \Sigma^1 = \{a, b\} \cdot \{a, b\}$
 $= \{\underline{aa}, \underline{ab}, \underline{ba}, \underline{bb}\}$

$$\underline{\Sigma^3} = \Sigma^2 \cdot \Sigma^1 = \{\underline{aaa}, \underline{aba}, \underline{baa}, \underline{bba}, \underline{aab}, \underline{abb}, \underline{bab}, \underline{bbb}\}$$

$$\Sigma^4 = \Sigma^3 \cdot \Sigma^1 = \{ \dots \}$$

$$\Sigma^m \rightarrow \{ \} m$$

$$\underline{\Sigma^*} = \{ \} \Sigma$$

$$\underline{\Sigma^*} = \Sigma^0 \cup \Sigma^1 \cup \Sigma^2 \dots$$

English

$$\underline{\Sigma} = \{a, \dots, z\}$$

$$\underline{\Sigma^*} = \{aa, ab, \dots, z z\}$$

a, an,
bat eat

6. KLEENE CLOSURE

- Let Σ be an alphabet. Then “Kleene Closure Σ^* ” denotes the set of all strings (including ϵ , empty string) over the alphabet Σ .

Examples

- If $\Sigma = \{a\}$ then $\Sigma^* = \{\epsilon, a, aa, aaa, \dots\}$ i.e.

$$\Sigma^0 = \{\epsilon\}$$

$$\Sigma^1 = \{a\}$$

$$\Sigma^2 = \{aa\}$$

- If $\Sigma = \{0, 1\}$ then $\Sigma^* = \{\epsilon, 0, 1, 00, 01, 10, \dots\}$
- If $\Sigma = \{0\}$ then $\Sigma^* = \{\epsilon, 0, 00, 000, \dots\}$
- $\therefore \Sigma^* = \Sigma^0 \cup \Sigma^1 \cup \Sigma^2 \cup \dots$

$$\Sigma^* = \bigcup_{i=0}^{\infty} \Sigma^i$$

6. KLEENE CLOSURE

- $\Sigma = \{a,b\}$
- $\Sigma^* = ?$

- $\Sigma = \{0,1,2\}$
- $\Sigma^* = ?$

$$\Sigma^0 = \{\epsilon\}$$

$$\Sigma^1 = \{0, 1, 2\}$$

$$\Sigma^2 = \{00, 01, 02, 10, 20, \dots\}$$

$$\Sigma^3 = \{000, 111, 222, \dots\}$$

$$\Sigma^* = \{\epsilon, 0, 1, 2, 00, \dots, 000, \dots\}$$

Σ^* Set of all strings over the alphabet Σ

7. SUBSTRING

- A string v appears within another string w ($w=uv$) is called “substring of w .” If $w=uv$, then substrings u & v are said to be prefix and suffix of w respectively.

Examples

- $w=abab$
Substring = { a , ab , abb , ba , bab ,...}
- $w = 123$
Prefixes = { ϵ , 1 , 12 , 123 }
Suffixes = { ϵ , 3 , 23 , 123 }
- $w = abab$
Prefixes = { ϵ , a , ab , abb , $abba$, $abab$ }
Suffixes = { ϵ , b , ab , bab , $bbab$, $abab$ }

8. PALINDROME

- A palindrome is a string, which is same whether written forward (or) backward.

- **Example**

madam, malayalam, noon, nun, 121.

- If the length of a palindrome is even, then it can be obtained by concatenation of a string and its reverse.

- **Example**

If $u = 01$ $u^R = \underline{10}$.

$w = 0$

uv^R

then even palindrome = 0110 ←

9. PROPERTIES OF STRING OPERATIONS

- Concatenation is associative ; that is for all strings u, v and w ,
 $(uv) w = u (vw)$
- If u and v are strings, then the length of their concatenation is the sum of the individual lengths, i.e.,

$$|uv| = \underline{|u|} + |v|.$$

Example

$$x = \underline{abc} \quad y = 123 \quad xy = abc123$$

$$|xy| = 6 \quad |x| = 3 \quad |y| = 3$$

$$\text{hence } |xy| = |x| + |y|$$

LANGUAGES

- A language is a set of strings which are made up of characters from a specified alphabet, or set of symbols.
- A set of strings all of which are chosen from some Σ^* , where Σ is a particular alphabet, is called a language.
- If Σ is an alphabet, and $\underline{L} \subseteq \Sigma^*$, then L is a language over Σ^* .

Examples

^L
English $\rightarrow \Sigma = \{a, b, c, \dots, z\}$

Binary strings : $\{0, 1, 01, 10, 0101, \dots\} \rightarrow \Sigma = \{0, 1\}$

$\Sigma^* = \{\epsilon, a, b, aa, ab, \dots\} \rightarrow \Sigma = \{a, b\}$

$\Sigma = \{0, 1\}$ set of all strings which ends with 000 $\Sigma = \{0, 1\}$
 $L = \{000, 1000, 0000, 11000, 10000, 010000, \dots\}$

$L \leftarrow$ meaningful strings.
 Σ
 $L \subseteq \Sigma^*$

$L = \{00, 11, 0000, 1010, 0101, 1100, 0011, \dots\}$

LANGUAGES

Notations

$$L = \{\varepsilon\}$$

- $\{\lambda\}$ (or) $\{\varepsilon\} \rightarrow$ Empty string (or) Null string language.

It is a language over every alphabet and it contains exactly one string ε (or) λ .

$$\underline{L = \{\lambda\}}$$

- ϕ : Empty language

It contains no strings.

$$\{\varepsilon\} \neq \{\lambda\}$$

- Σ^* : Universal language

It contains all (finite) string over the alphabet Σ .

Note

- $\phi \neq \{\lambda\}$ ie ϕ has no string where as $\{\varepsilon\}$ (or) $\{\lambda\}$ has one string ε (or) λ .

OPERATIONS ON LANGUAGES

A. PRODUCT (OR) CONCATENATION

- $L_1 \cdot L_2 = L_1 L_2 = \{xy : x \in L_1, y \in L_2\}$
- ie., the concatenation of two languages L_1 and L_2 are set of all strings contained by
- concatenating any element of L_1 with any element of L_2 .

A. PRODUCT (OR) CONCATENATION

- $L1 = \{a, b, aa, bb\}$
- $L2 = \{0, 1, 00, 11, 01, 10\}$
- $L1.L2 = ?$

$\{a0, b0, aa0, bb0, a1, b1, bb1, \dots\}$

B. REVERSAL

- The reverse of a language is the set of all string reversals. ie., $L^R = \{w^R : w \in L\}$

- $L1 = \{a,b,ab,ba,bba,baba,\}$

w^R

$w = 100$

$w^R = 001$

- $L2 = \{0,1,01,10,1011,0101\}$

- $L1^R = ?$

$\{a,b,ba,ab,abb,abab\}$

- $L2^R$

$\{0,1,10,01,1101,1010\}$

C. POWER

- For a given language L , $L^0 = \{\lambda\}$
- We define L^n as L concatenated itself n times

$$\text{ie } \underline{L^0} = \underline{\{x\}}$$

$$\underline{L^1} = L$$

$$\underline{L^K} = L . \underline{L^{K-1}}$$

(or)

$$L^K = \{x_1 \dots x_K : x_i \in L\} \text{ where } i \text{ ranges from } 1 \text{ to } K.$$

Σ^*
 Σ^1
 Σ^0
 Σ^1

D. KLEENE STAR (OR) STAR CLOSURE

- For a language L,

$$L^* = \bigcup_{i=0}^{\infty} L^i = \underline{\underline{L^0 \cup L^1 \cup L^2 \cup \dots}}$$

- $L_1 = \{a, b, ab, ba, bba, baba, \dots\}$**
- $L_1^* = ?$**

$$L^0 = \{\epsilon\}$$

$$L^1 = L$$

$$L^2 = L \cdot L = \{ \dots \}$$

$$L^3 = L^2 \cdot L = \{ \dots \} \quad L^*$$

E. KLEENE PLUS (OR) POSITIVE CLOSURE

- $L^+ = \bigcup_{i=1}^{\infty} L^i = L^1 \cup L^2 \cup \dots$

- $L1 = \{a, b, ab, ba, bba, baba, \dots\}$

- $L1^+ = ?$

$$\begin{aligned} L &= \{a, b\} \\ L^+ &= \{a, b, ab, ba, bba, baba, \dots\} \end{aligned}$$

F. UNION

- The union of L_1 and L_2 denoted by $L_1 \cup L_2$ is
- $L_1 \cup L_2 = \{w: w \in L_1 \text{ or } w \in L_2\}$
- $L_1 = \{a,b,ab,ba,bba,baba,\}$
- $L_2 = \{0,1,01,10,1011,0101\}$
- $L_1 \cup L_2 = ?$

$$L_1 \cap L_2 = \{\} = \emptyset$$

G. INTERSECTION

- The intersection of L_1 and L_2 denoted by $L_1 \cap L_2$ is
- $L_1 \cap L_2 = \{w : w \in L_1 \text{ and } w \in L_2\}$
- **$L_1 = \{a,b,ab,ba,bba,baba,\}$**
- **$L_2 = \{0,1,01,10,1011,0101\}$**
- **$L_3 = \{0,1,a,b\}$**
- **$L_1 \cap L_2 = ?$**
- **$L_1 \cap L_3 = ?$**

GRAPHS

- A graph, denoted by $G = (V, E)$ consists of a finite set of vertices (or) nodes V and a set E , a pair of vertices called edges.
- *A path in a graph is a sequence of vertices $v_1, v_2, v_3, \dots, v_k, k \geq 1$ such that there is an edge (v_i, v_{i+1}) for each $i, 1 \leq i < k$.*
- The length of the path is $k-1$.
- *If $v_1 = v_k$, then the path is said to be cycle (because starting and ending at same vertex).*

TREES

- *A tree (strictly speaking ordered, directed tree) is a digraph satisfying following properties:*
 - (i) There is one vertex called the root, of the tree which is distinguished from all other vertices and the root has no predecessors.
 - (ii) There is a directed path from the root to every other vertex.
 - (iii) Every vertex except the root has exactly one predecessor.
 - (iv) The successors of each vertex are ordered from left to right.

TEST YOUR KNOWLEDGE

- For any languages L_1, L_2, L over $\Sigma \neq \emptyset$,
 $(L_1 \cup L_2) \cap L = (L_1 \cap L) \cup (L_2 \cap L)$

Justify your answer

- For any language L over an alphabet Σ ,
 $L^+ = L \cup L^*$

True or false

SUMMARY

- Introduction about basic mathematical notations - alphabet, strings, languages
- Discussion about different operations on strings
- Languages and operations on languages
- Definition on graph, trees

LEARNING OUTCOME

On successful completion of this topic, the student will be able to:

- Understand the need of basic mathematical notations (K2)

REFERENCE

- Hopcroft J.E., Motwani R. and Ullman J.D, “Introduction to Automata Theory, Languages and Computations”, Second Edition, Pearson Education, 2008